

Subodh Kathayat

862-214-5213 | kathayatsubodh20@gmail.com | [linkedin/subodh-kathayat](https://www.linkedin.com/in/subodh-kathayat) | github.com/subodh | [Portfolio](#)

EDUCATION

New Jersey Institute of Technology

Bachelor of Science in Computer Science

Involvement: Peer Mentor at CDS, Vice President of NEPSA, E-board member of Badminton Club, Member of ACM

Expected May 2027

Newark, NJ

SKILLS

Programming: Python, MySQL, Swift, C++

Developer Tools: Git/Github, Xcode, VS Code, Generative AI, Figma (UI/UX), Adobe

Certifications: AWS Cloud Practitioner Certification (In-progress)

EXPERIENCE

Data Management Intern, CDS

New Jersey Institute of Technology

Feb. 2025 – Present

Newark, NJ

- Incoming Spring data management intern (Python, MySQL, PySide, PowerBI)

Undergrad Research Assistant

New Jersey Institute of Technology

Sept. 2024 – Feb. 2025

Newark, NJ

- Designing a swarm robotic localization system using **Python3** (Pymunk, Pygame) to optimize robot positioning algorithms, resulting in a 25% reduction in localization error
- Collaborating with teammates to apply **data analysis** and visualization techniques to real-world robot **datasets**

Software Engineering Fellow

Headstarter AI

July 2024 – Sep. 2024

Remote

- Built backends for 3 AI apps using **Python, OpenAI, and Stripe API** with 98% accuracy seen by 1,000 users
- Developed projects from design to deployment leading 2+ engineering fellows using MVC design patterns
- Coached by Amazon, Bloomberg, and Capital One engineers on Agile, CI/CD, Git, and microservice patterns

PROJECTS

Echoes (In-Progress) | Swift, SwiftUI, AVFoundation, Core ML, Figma

- Building an iOS app with Swift and Core ML for seamless communication via **text-to-speech** and speech-to-text
- Implemented a locally trained AI model for on-device document processing and intelligent responses

Neptou - The Tourism Bud | Swift, Google Maps API, Figma

- Engineered Neptou's mobile architecture using Swift and Google Maps API, implementing scalable location-based services and validating its impact on the tourism sector
- Earned national recognition as a Top 5 Finalist in the **ICT Award 2022** (Nepal's most prestigious IT award) and secured 4th place in Code for Change 2022, the country's largest hackathon, competing against 400 participants

Ragnarok AI (Hackathon NJIT 2024) | Next.js, LangChain, OpenAI, Vercel, Github

- Programmed an AI-powered chatbot for real-time Badminton news updates using **LangChain**, Next.js, Vercel, and OpenAI, enhancing response accuracy with web scraping and vector embeddings
- Implemented a Retrieval-Augmented Generation (**RAG**) system that helped in real-time info

Terminal Asteroids game | C++, Git

- Created a console-based Asteroids game in C++, leveraging OOP and physics to create realistic gameplay
- Designed a **toroidal mapping system** and optimized collision detection, enhancing gameplay responsiveness

LEADERSHIP

Founder & President

Aasha Ka Paila, Non-Profit Organization

2020 – Present

Nepal

- Founded and lead a non-profit organization focused on supporting underprivileged families in Nepal, providing resources such as food, clothes, and education.
- Coordinated community outreach programs that helped over 700 families in the past 4 years.
- Managed a team of 25+ volunteers, overseeing project planning, fundraising, and community engagement efforts.
- Organized fundraising events and partnerships, raising over \$5,000 for the organization's initiatives.